Evaluation runs: 2pc (a1-commit)

generated 2026-11-03 from brain-data e6c29d7dd 2026-11-03 20:14:23 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): correct: expected conforms (n=1 is refused: the program never spawns visible p1); control-reverse: expected conforms (n=1 is refused: the program never spawns visible p1); eager: expected violation (n=1 is refused: the program never spawns visible p1); early-abort: expected violation (n=1 is refused: the program never spawns visible p1); silent: expected violation (n=1 is refused: the program never spawns visible p1). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair control-reverse, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,1	16	8	conforms	0.0	4	1 h, 250 GB	4	29	brain03, 2026-11-03
2,2	256	64	conforms	0.1	4	1 h, 250 GB	16	1658	brain03, 2026-11-03
3,1	16	48	conforms	0.0	4	1 h, 250 GB	5	38	brain03, 2026-11-03
4,2	256	147456	conforms	4.2	6	1 h, 250 GB	25	2353	brain03, 2026-11-03
5,3			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,1	16	8	conforms	0.0	4	1 h, 250 GB	4	30	brain03, 2026-11-03
2,2	256	64	conforms	0.1	4	1 h, 250 GB	16	1847	brain03, 2026-11-03
3,1	16	48	conforms	0.0	4	1 h, 250 GB	5	30	brain03, 2026-11-03
4,2	256	147456	conforms	4.2	8	1 h, 250 GB	25	3219	brain03, 2026-11-03
5,3			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair eager, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,1	16	8	5	violation	0.0	4	1 h, 250 GB	8	101	brain03, 2026-11-03
2,2	256	64	55	violation	0.7	2	1 h, 250 GB	64	14966	brain03, 2026-11-03
3,1	16	48	42	violation	0.1	4	1 h, 250 GB	45	698	brain03, 2026-11-03
4,2	256	147456	146880	violation	502.1	19	1 h, 250 GB	146889	37601984	brain03, 2026-11-03
5,3				🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair early-abort, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,1	16	8	4	violation	0.0	4	1 h, 250 GB	7	83	brain03, 2026-11-03
2,2	256	64	48	violation	0.6	4	1 h, 250 GB	57	13455	brain03, 2026-11-03

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
3,1	16	48	32	violation	0.1	4	1 h, 250 GB	36	545	brain03, 2026-11-03
4,2	256	147456	139712	violation	462.9	19	1 h, 250 GB	139728	35768350	brain03, 2026-11-03
5,3				(1 h)			1 h, 250 GB			brain03, 2026-11-01

stateless: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
2,1	16	8	8	violation	0.0	4	1 h, 250 GB	8	128	brain03, 2026-11-03
2,2	256	64	64	violation	0.8	4	1 h, 250 GB	64	16384	brain03, 2026-11-03
3,1	16	48	48	violation	0.1	4	1 h, 250 GB	48	768	brain03, 2026-11-03
4,2	256	147456	147456	violation	384.1	4	1 h, 250 GB	147456	37748736	brain03, 2026-11-01
5,3				(1 h)			1 h, 250 GB			brain03, 2026-11-01

Not run (stateless). planned but never scheduled (grid points beyond the ladder): control-reverse 2 3 fifo, correct 2 3 fifo, early-abort 2 3 fifo, silent 2 3 fifo, eager 2 3 fifo, silent 2 4 fifo, early-abort 2 4 fifo, eager 2 4 fifo, correct 2 4 fifo, control-reverse 2 4 fifo, eager 2 5 fifo, correct 2 5 fifo, silent 2 5 fifo, early-abort 2 5 fifo, control-reverse 2 5 fifo, early-abort 3 2 fifo, silent 3 2 fifo, eager 3 2 fifo, correct 3 2 fifo, control-reverse 3 2 fifo, early-abort 3 3 fifo, eager 3 3 fifo, control-reverse 3 3 fifo, correct 3 3 fifo, … (95 instances in all).

skipped: a smaller size had reached its limit: early-abort 6 4 fifo, eager 6 4 fifo, control-reverse 6 4 fifo, correct 6 4 fifo, silent 6 4 fifo, correct 7 5 fifo, silent 7 5 fifo, control-reverse 7 5 fifo, early-abort 7 5 fifo, eager 7 5 fifo, correct 8 6 fifo, silent 8 6 fifo, eager 8 6 fifo, early-abort 8 6 fifo, control-reverse 8 6 fifo, control-reverse 9 7 fifo, silent 9 7 fifo, correct 9 7 fifo, eager 9 7 fifo, early-abort 9 7 fifo.

stateful

stateful: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,1	16	8	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-03
2,2	256	64	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-03
3,1	16	48	conforms	0.3	2	1 h, 250 GB	brain02, 2026-11-03
4,2	256	147456	conforms	4.7	2	1 h, 250 GB	brain02, 2026-11-03
5,3			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
2,1	16	8	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-03
2,2	256	64	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,1	16	48	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-03
4,2	256	147456	conforms	4.5	4	1 h, 250 GB	brain02, 2026-11-03
5,3			(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,1	16	8	5	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2	256	64	55	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,1	16	48	42	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4,2	256	147456	146880	violation	5.1	8	1 h, 250 GB	brain02, 2026-11-03
5,3				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,1	16	8	4	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-03
2,2	256	64	48	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-03
3,1	16	48	32	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4,2	256	147456	139712	violation	4.7	6	1 h, 250 GB	brain02, 2026-11-03
5,3				(1 h)			1 h, 250 GB	brain02, 2026-11-03

stateful: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
2,1	16	8	8	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2,2	256	64	64	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3,1	16	48	48	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4,2	256	147456	147456	violation	4.5	4	1 h, 250 GB	brain02, 2026-11-03
5,3				(1 h)			1 h, 250 GB	brain02, 2026-11-01

Not run (stateful). planned but never scheduled (grid points beyond the ladder): control-reverse 2 3 fifo, correct 2 3 fifo, early-abort 2 3 fifo, silent 2 3 fifo, eager 2 3 fifo, silent 2 4 fifo, early-abort 2 4 fifo, eager 2 4 fifo, correct 2 4 fifo, control-reverse 2 4 fifo, eager 2 5 fifo, correct 2 5 fifo, silent 2 5 fifo, early-abort 2 5 fifo, control-reverse 2 5 fifo, early-abort 3 2 fifo, silent 3 2 fifo, eager 3 2 fifo, correct 3 2 fifo, control-reverse 3 2 fifo, early-abort 3 3 fifo, eager 3 3 fifo, control-reverse 3 3 fifo, correct 3 3 fifo, … (95 instances in all).

skipped: a smaller size had reached its limit: early-abort 6 4 fifo, eager 6 4 fifo, control-reverse 6 4 fifo, correct 6 4 fifo, silent 6 4 fifo, correct 7 5 fifo, silent 7 5 fifo, control-reverse 7 5 fifo, early-abort 7 5 fifo, eager 7 5 fifo, correct 8 6 fifo, silent 8 6 fifo, eager 8 6 fifo, early-abort 8 6 fifo, control-reverse 8 6 fifo, control-reverse 9 7 fifo, silent 9 7 fifo, correct 9 7 fifo, eager 9 7 fifo, early-abort 9 7 fifo.

offline

offline: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
2,2	256	64	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,1	16	48	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4,2	256	147456	conforms	4.3	4	10 min, 16 GB	0.1	4.3	0.0	in-memory	brain05, 2026-11-02

offline: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
2,2	256	64	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,1	16	48	conforms	0.1	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
4,2	256	147456	conforms	4.2	4	1 h, 250 GB	0.1	4.2	0.0	merge	brain05, 2026-11-03

offline: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	5	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
2,2	256	64	55	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain05, 2026-11-02
3,1	16	48	25	violation	0.1	4	1 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
4,2	256	147456	1015	violation	4.1	4	1 h, 250 GB	0.1	4.1	0.0	merge	brain05, 2026-11-03

offline: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	4	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
2,2	256	64	40	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,1	16	48	12	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4,2	256	147456	240	violation	4.3	4	10 min, 16 GB	0.1	4.3	0.0	in-memory	brain05, 2026-11-02

offline: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	4	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
2,2	256	64	16	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
3,1	16	48	5	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4,2	256	147456	25	violation	27.5	4	10 min, 16 GB	0.1	27.5	0.0	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): control-reverse 2 3 fifo, correct 2 3 fifo, early-abort 2 3 fifo, silent 2 3 fifo, eager 2 3 fifo, silent 2 4 fifo, early-abort 2 4 fifo, eager 2 4 fifo, correct 2 4 fifo, control-reverse 2 4 fifo, eager 2 5 fifo, correct 2 5 fifo, silent 2 5 fifo, early-abort 2 5 fifo, control-reverse 2 5 fifo, early-abort 3 2 fifo, silent 3 2 fifo, eager 3 2 fifo, correct 3 2 fifo, control-reverse 3 2 fifo, early-abort 3 3 fifo, eager 3 3 fifo, control-reverse 3 3 fifo, correct 3 3 fifo, … (100 instances in all).

skipped: a smaller size had reached its limit: early-abort 5 3 fifo, eager 5 3 fifo, silent 5 3 fifo, correct 5 3 fifo, control-reverse 5 3 fifo, early-abort 6 4 fifo, eager 6 4 fifo, control-reverse 6 4 fifo, correct 6 4 fifo, silent 6 4 fifo, correct 7 5 fifo, silent 7 5 fifo, control-reverse 7 5 fifo, early-abort 7 5 fifo, eager 7 5 fifo, correct 8 6 fifo, silent 8 6 fifo, eager 8 6 fifo, early-abort 8 6 fifo, control-reverse 8 6 fifo, control-reverse 9 7 fifo, silent 9 7 fifo, correct 9 7 fifo, eager 9 7 fifo, … (25 instances in all).

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	conforms	3.1	174	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	conforms	3.6	166	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	conforms	3.3	164	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	conforms	3.1	162	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	conforms	3.4	176	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	conforms	3.4	162	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	5	violation	3.1	156	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	55	violation	3.4	160	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	25	violation	3.3	167	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	4	violation	3.2	174	10 min, 16 GB	0.0	3.2	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	40	violation	3.4	175	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	12	violation	3.4	168	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-01

cross offline, Java implementation against the Rust specification: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	4	violation	3.1	152	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	16	violation	3.4	170	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	5	violation	3.2	167	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	conforms	3.2	174	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	conforms	4.0	174	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	conforms	3.1	168	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	conforms	3.2	174	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	conforms	4.0	174	10 min, 16 GB	4.0	0.2	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	conforms	3.1	168	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	5	violation	3.2	174	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	55	violation	4.0	174	10 min, 16 GB	4.0	0.2	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	25	violation	3.1	168	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	4	violation	3.2	174	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-01
2,2	256	64	40	violation	4.0	174	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	12	violation	3.1	168	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01

cross offline, Rust implementation against the Java specification: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	16	8	4	violation	3.2	174	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-01

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,2	256	64	16	violation	4.0	174	10 min, 16 GB	4.0	0.1	0.0	in-memory	brain01, 2026-11-01
3,1	16	48	5	violation	3.1	168	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-01

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	16	8	conforms	0.7	149	1 h, 250 GB		brain05, 2026-11-02
2,2	256	64	conforms	1.0	148	1 h, 250 GB		brain05, 2026-11-02
3,1	16	48	conforms	0.8	146	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	16	8	conforms	0.6	167	1 h, 250 GB		brain05, 2026-11-02
2,2	256	64	conforms	1.0	152	1 h, 250 GB		brain05, 2026-11-02
3,1	16	48	conforms	0.8	148	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	16	8	5	violation	0.7	146	1 h, 250 GB	46	brain05, 2026-11-02
2,2	256	64	55	violation	1.1	156	1 h, 250 GB	52	brain05, 2026-11-02
3,1	16	48	42	violation	0.8	150	1 h, 250 GB	47	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	16	8	4	violation	0.6	148	1 h, 250 GB	88	brain05, 2026-11-02
2,2	256	64	48	violation	1.0	156	1 h, 250 GB	124	brain05, 2026-11-02
3,1	16	48	32	violation	0.9	148	1 h, 250 GB	121	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	16	8	8	violation	0.6	146	1 h, 250 GB	41	brain05, 2026-11-02
2,2	256	64	64	violation	1.1	156	1 h, 250 GB	65	brain05, 2026-11-02
3,1	16	48	48	violation	0.8	167	1 h, 250 GB	53	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): eager 2 1 fifo, early-abort 2 1 fifo, silent 2 1 fifo, correct 2 1 fifo, control-reverse 2 1 fifo, eager 2 2 fifo, control-reverse 2 2 fifo, silent 2 2 fifo, early-abort 2 2 fifo, correct 2 2 fifo, early-abort 3 1 fifo, silent 3 1 fifo, correct 3 1 fifo, control-reverse 3 1 fifo, eager 3 1 fifo.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	16	8	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2	256	64	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3,1	16	48	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	16	8	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
2,2	256	64	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01
3,1	16	48	conforms	0.0	4	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	16	8	5	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,2	256	64	55	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3,1	16	48	42	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	16	8	4	violation	0.0	4	1 h, 250 GB	1	brain04, 2026-11-01
2,2	256	64	48	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3,1	16	48	32	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	16	8	8	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
2,2	256	64	64	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01
3,1	16	48	48	violation	0.0	4	1 h, 250 GB	2	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). left out by the table rules (ok): eager 2 1 fifo, early-abort 2 1 fifo, silent 2 1 fifo, correct 2 1 fifo, control-reverse 2 1 fifo, eager 2 2 fifo, control-reverse 2 2 fifo, silent 2 2 fifo, early-abort 2 2 fifo, correct 2 2 fifo, early-abort 3 1 fifo, silent 3 1 fifo, correct 3 1 fifo, control-reverse 3 1 fifo, eager 3 1 fifo.

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.2	154	10 min, 16 GB	3.1	0.5	0.1	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	3.6	167	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms	3.8	160	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair control-reverse, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms*	9.4	160	10 min, 16 GB	9.4	0.5	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms*	3.6	164	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms*	3.6	163	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	9.8	160	10 min, 16 GB	9.4	0.1	0.4	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	5.0	160	10 min, 250 GB	4.8	0.2	0.1	merge	brain02, 2026-11-03
3,1	48	48	conforms	4.9	175	10 min, 250 GB	4.9	0.2	0.0	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair eager, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.6	154	10 min, 16 GB	3.3	0.1	0.2	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	4.3	157	10 min, 250 GB	4.3	0.2	0.0	merge	brain02, 2026-11-03
3,1	48	48	conforms	8.2	158	10 min, 250 GB	8.2	0.2	0.0	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair early-abort, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.3	160	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	3.6	154	10 min, 16 GB	3.6	0.4	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms	3.5	164	10 min, 16 GB	3.4	0.5	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair silent, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.7	169	10 min, 16 GB	3.7	0.5	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	3.8	163	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms	3.4	156	10 min, 16 GB	3.3	0.1	0.1	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair eager, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	5	violation	9.4	160	10 min, 16 GB	9.4	0.1	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	55	violation	4.8	160	10 min, 250 GB	4.8	0.2	0.0	merge	brain02, 2026-11-03
3,1	48	48	25	violation	4.9	175	10 min, 250 GB	4.9	0.2	0.0	merge	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	4	violation	9.4	160	10 min, 16 GB	9.4	0.1	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	40	violation	3.6	164	10 min, 16 GB	3.6	0.4	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	12	violation	3.6	163	10 min, 16 GB	3.6	0.5	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	4	violation	9.4	160	10 min, 16 GB	9.4	0.5	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	16	violation	3.6	164	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	5	violation	3.6	163	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.1	154	10 min, 16 GB	0.5	3.1	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	3.6	167	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms	3.8	160	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms*	3.1	154	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms*	3.6	167	10 min, 16 GB	0.2	3.6	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms*	3.8	160	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **correct**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	9.4	160	10 min, 16 GB	0.1	9.4	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	4.8	160	10 min, 250 GB	0.2	4.8	0.0	merge	brain02, 2026-11-03
3,1	48	48	conforms	4.9	175	10 min, 250 GB	0.2	4.9	0.0	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **eager**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.3	154	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	4.3	157	10 min, 250 GB	0.2	4.3	0.0	merge	brain02, 2026-11-03
3,1	48	48	conforms	8.2	158	10 min, 250 GB	0.2	8.2	0.0	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **early-abort**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.3	160	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	3.6	154	10 min, 16 GB	0.4	3.6	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms	3.4	164	10 min, 16 GB	0.5	3.4	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **silent**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	conforms	3.7	169	10 min, 16 GB	0.5	3.7	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	conforms	3.8	163	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	conforms	3.3	156	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **eager**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	5	violation	3.3	154	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	55	violation	4.3	157	10 min, 250 GB	0.2	4.3	0.0	merge	brain02, 2026-11-03
3,1	48	48	25	violation	8.2	158	10 min, 250 GB	0.2	8.2	0.0	merge	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **early-abort**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	4	violation	3.3	160	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	40	violation	3.6	154	10 min, 16 GB	0.2	3.6	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	12	violation	3.4	164	10 min, 16 GB	0.2	3.4	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **silent**, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
2,1	8	8	4	violation	3.7	169	10 min, 16 GB	0.1	3.7	0.0	in-memory	brain02, 2026-11-03
2,2	64	64	16	violation	3.8	163	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain02, 2026-11-03
3,1	48	48	5	violation	3.3	156	10 min, 16 GB	0.2	3.3	0.0	in-memory	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **control-reverse**, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	3.3	164	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	3.8	162	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	3.6	158	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair control-reverse, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms*	3.5	162	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms*	3.8	167	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms*	3.5	163	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	4.1	160	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	5.1	156	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	4.8	155	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair eager, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	3.9	158	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	5.2	161	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	5.0	158	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair early-abort, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	3.3	150	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	4.0	160	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	3.5	150	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair silent, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	3.4	164	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	4.0	162	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	3.5	154	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair eager, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	8	8	5	violation	3.6	164	1 h, 250 GB	132	brain02, 2026-11-03
2,2	64	64	55	violation	5.0	155	1 h, 250 GB	128	brain02, 2026-11-03
3,1	48	48	42	violation	4.5	150	1 h, 250 GB	86	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair early-abort, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	8	8	4	violation	3.5	164	1 h, 250 GB	2260	brain02, 2026-11-03
2,2	64	64	48	violation	3.8	157	1 h, 250 GB	2340	brain02, 2026-11-03
3,1	48	48	32	violation	3.8	158	1 h, 250 GB	2196	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair silent, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	8	8	8	violation	3.5	166	1 h, 250 GB	64	brain02, 2026-11-03
2,2	64	64	64	violation	3.8	157	1 h, 250 GB	85	brain02, 2026-11-03
3,1	48	48	48	violation	3.8	153	1 h, 250 GB	69	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair control-reverse, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair control-reverse, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms*	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms*	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms*	0.0	2	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair correct, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	0.2	4	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair eager, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair early-abort, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair silent, model fifo, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
2,1	8	8	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
2,2	64	64	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3,1	48	48	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair eager, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	8	8	5	violation	0.0	4	1 h, 250 GB	1	brain02, 2026-11-03
2,2	64	64	55	violation	0.0	4	1 h, 250 GB	1	brain02, 2026-11-03
3,1	48	48	42	violation	0.0	4	1 h, 250 GB	1	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair early-abort, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	8	8	4	violation	0.0	4	1 h, 250 GB	1	brain02, 2026-11-03
2,2	64	64	48	violation	0.0	4	1 h, 250 GB	2	brain02, 2026-11-03
3,1	48	48	32	violation	0.0	2	1 h, 250 GB	1	brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair silent, model fifo, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2,1	8	8	8	violation	0.0	4	1 h, 250 GB	1	brain02, 2026-11-03
2,2	64	64	64	violation	0.0	2	1 h, 250 GB	1	brain02, 2026-11-03
3,1	48	48	48	violation	0.0	4	1 h, 250 GB	1	brain02, 2026-11-03